

JEE MAIN REVISION MASTER: IONIC EQUILIBRIUM

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 140 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	pH Calculation of Acids, Bases & Salts	101 PYQs	pH & Hydrolysis
Very High	Solubility Product (Ksp) & Common Ion Effect	45 PYQs	Solubility

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
pH Calculation of Acids, Bases & Salts	$\text{pH} = \text{p}K_a + \log \frac{[\text{conjugate base}]}{[\text{acid}]}, \quad \text{Salt of WA-SB: } \text{pH} = 7 + \frac{1}{2} \text{p}K_a + \frac{1}{2} \log c$
Solubility Product (Ksp) & Common Ion Effect	$A_x B_y \rightleftharpoons x A^{y+} + y B^{x-} \implies K_{sp} = x^x y^y s^{x+y}, \quad Q_{sp} > K_{sp} \implies \text{Precipitation occurs}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 pH Calculation of Acids, Bases & Salts (Very High Yield — 101 PYQs)

Core Principle: Self-ionization of water ($K_w = 10^{-14}$ at 25°C), Ostwald dilution law, buffer solutions, and salt hydrolysis formulas.

■ **Caution / JEE Pitfall:** pH of 10^{-8} M HCl is NOT 8 (cannot be basic!); including water's auto-ionization $[H^+] = 10^{-8} + 10^{-7}$ implies $\text{pH} \approx 6.79$.

■ **Speed Trick:** Buffer capacity is maximum when $[\text{salt}] = [\text{acid}] \implies \text{pH} = \text{p}K_a$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given is a concentrated solution of a weak electrolyte $A_x B_y$ of concentration ' c ' and dissociation constant ' K '. The degree of ..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "20 mL of a solution of acetic acid required 28.4 mL of 0.1 M NaOH for its neutralization. A solution (X) was prepared by mixing 20 mL of the..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Consider a weak base ' B ' of $\text{p}K_b = 5.699$. ' x ' mL of 0.02 M HCl and ' y ' mL of 0.02 M weak base ' B ' are mixed to make 100 mL of a..." [Ans: N/A]

3.2 Solubility Product (K_{sp}) & Common Ion Effect (Very High Yield — 45 PYQs)

Core Principle: Calculation of molar solubility s from K_{sp} , ionic product comparison, and suppression of solubility via common ion effect.

■ **Caution / JEE Pitfall:** For Ag_2CrO_4 (A₂B type): $K_{sp} = (2s)^2(s) = 4s^3$ implies $s = \sqrt[3]{K_{sp}/4}$, NOT s^2 !

■ **Speed Trick:** Common ion effect ALWAYS suppresses the solubility of a sparingly soluble salt (e.g. AgCl is less soluble in NaCl than pure water).

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (Numerical):** "The first and second ionization constants of H_2X are 2.5×10^{-8} and 1.0×10^{-13} respectively. The concentration of X^{2-} ..." [Ans: 100]
- **[2026 Morning Shift] Q#2 (MCQ):** " M_3A_2 is a sparingly soluble salt of molar mass $y \text{ g mol}^{-1}$ and solubility $x \text{ g L}^{-1}$. The ratio of the molar concentration of the ..." [Ans: A]
- **[2026 Morning Shift] Q#4 (MCQ):** "Given below are two statements : Statement I : Sodium dichromate and potassium dichromate are classified as primary standards in titrimetric..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 21st January Evening Shift (Q#1)

The first and second ionization constants of H_2X are 2.5×10^{-8} and 1.0×10^{-13} respectively. The concentration of X^{2-} in 0.1M H_2X solution is _____ $\times 10^{-15} \text{ M}$. (Nearest Integer)

Correct Answer: 100

Solution Analysis: For the diprotic acid H_2X (total concentration $C=0.1\text{M}$): $K_{a1}=2.5 \times 10^{-8}$, $K_{a2}=1.0 \times 10^{-13}$
Find $[\text{H}^+]$ (second dissociation is negligible for $[\text{H}^+]$): $[\text{H}^+] \approx \sqrt{(K_{a1}C)} = \sqrt{(2.5 \times 10^{-8})(0.1)} = \sqrt{2.5 \times 10^{-9}} = 5.0 \times 10^{-5} \text{ M}$
Use diprotic distribution for $[\text{X}^{2-}]$: $[\text{X}^{2-}] = C \cdot \frac{K_{a1}K_{a2}}{[\text{H}^+]^2 + K_{a1} + K_{a1}K_{a2}/[\text{H}^+]}$
Compute key terms: $\frac{K_{a1}K_{a2}}{[\text{H}^+]^2} = \frac{(2.5 \times 10^{-8})(1.0 \times 10^{-13})}{(5 \times 10^{-5})^2} = \frac{2.5 \times 10^{-21}}{2.5 \times 10^{-9}} = 10^{-12}$

Benchmark Problem 2: JEE Main 2026 (Online) 22nd January Evening Shift (Q#2)

Which of the following mixture gives a buffer solution with $\text{pH}=9.25$? Given : $\text{p}K_b(\text{NH}_4\text{OH})=4.75$

- (A) 0.5M NH_4OH (0.2L)+0.2M HCl (0.5L)
- (B) 0.2M NH_4OH (0.5L)+0.1M HCl (0.5L) **[Correct]**
- (C) 0.2M NH_4OH (0.4L)+0.1M HCl (1L)
- (D) 0.4M NH_4OH (1L)+0.1M HCl (1L)

Correct Answer: B

Solution Analysis: For a basic buffer of NH_4OH and its salt NH_4Cl , NCERT uses: $\text{pOH} = \text{p}K_b + \log\left(\frac{[\text{salt}]}{[\text{base}]}\right)$
Given $\text{p}K_b(\text{NH}_4\text{OH})=4.75$ and required $\text{pH}=9.25$: $\text{pOH}=14-9.25=4.75$ So, $4.75=4.75+\log\left(\frac{[\text{salt}]}{[\text{base}]}\right) \Rightarrow \log\left(\frac{[\text{salt}]}{[\text{base}]}\right)=0 \Rightarrow \frac{[\text{salt}]}{[\text{base}]}=1$ So, we need equal amounts of salt and base after mixing. Reaction: $\text{NH}_4\text{OH} + \text{HCl} \rightarrow \text{NH}_4\text{Cl} + \text{H}_2\text{O}$ If initial moles are n_b (base) and n_a (acid), then after reaction (for buffer, base must be in excess): moles of salt